

Qwen3-1.7B Knowledge Distillation

Offline sequence-level knowledge distillation with automatically validated teacher-generated reasoning traces

74.68% -> 79.61% GSM8K exact match	+65 net additional correct	19.5% error reduction	~28% teacher gap closed
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Executive summary

This experiment tested whether a compact Qwen3-1.7B student could improve mathematical word-problem performance by learning from worked solutions produced by a substantially larger Qwen3-14B-AWQ teacher. The teacher generated granular, step-by-step GSM8K demonstrations; each generation was automatically validated against the GSM8K ground-truth final answer and screened for reasoning/format quality before SFT. The student was then fine-tuned with 4-bit QLoRA and evaluated on the untouched 1,319-example GSM8K test split.

The distilled student improved from 74.68% to 79.61% exact-match accuracy, increasing the number of correct answers from 985 to 1050. That is a +4.93-percentage-point gain and a 19.5% reduction in errors. The paired analysis is especially informative: 157 previously wrong answers became correct, while 92 previously correct answers regressed.

Key result

The gain is not only a formatting effect. Exact-match scoring depends on the extracted numerical answer, and it improved by 4.93 percentage points on held-out GSM8K test problems.

1. What we did

- Teacher: Qwen3-14B-AWQ generated complete worked solutions for GSM8K training problems.
- Reasoning design: demonstrations used granular steps with at most one arithmetic operation per step.
- Quality control: outputs were automatically validated against ground-truth answers and filtered for reasoning/format quality; 1,922/2,000 train and 485/500 validation demonstrations were accepted.
- Student: Qwen3-1.7B was fine-tuned for 3 epochs using 4-bit NF4 QLoRA.
- Evaluation: the base and distilled student were scored with the same evaluator on all 1,319 untouched GSM8K test examples.

2. Distillation pipeline and supervision

Offline sequence-level knowledge distillation with validated teacher data

Teacher solutions are generated and automatically validated before the fixed traces are used as supervised targets.

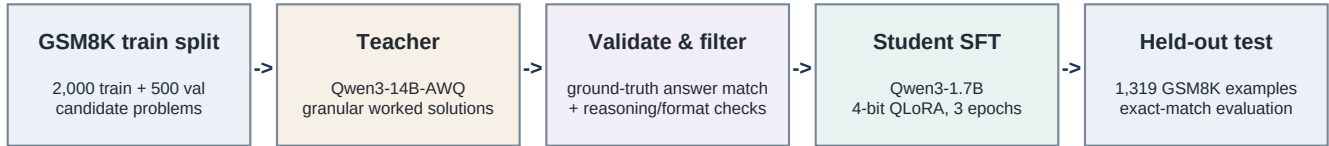


Figure 1. Teacher generation -> automatic validation/filtering -> QLoRA SFT -> held-out GSM8K evaluation.

Teacher-generated data and quality control

The teacher-generation stage started from 2,000 candidate training problems and 500 candidate validation problems from the GSM8K training split. Each output was automatically validated against the GSM8K ground-truth final answer and screened for required reasoning detail, tag/format structure, truncation, repetition, and related quality failures. The final SFT data contained 1,922 accepted training demonstrations (96.1%) and 485 accepted validation demonstrations (97.0%). Test problems were never used as student supervision.

Split	Candidates	Accepted	Acceptance rate
Train	2,000	1,922	96.1%
Validation	500	485	97.0%

Granular reasoning target

The teacher prompt was deliberately written for a smaller student. Multi-operation arithmetic was split into small, explicit transitions, and every calculating step was expected to state its numeric result. The intent was to provide dense supervision over intermediate quantities and operation order, rather than only a final number.

Illustrative target style

Preferred: $50 + 80 = 130$, then $130 + 60 = 190$. Avoid compressing both operations into $50 + 80 + 60 = 190$.

Method classification

This experiment uses offline sequence-level knowledge distillation with teacher-generated reasoning traces. Teacher solutions are generated and automatically validated before training, then fixed and used as supervised targets for Qwen3-1.7B through QLoRA SFT.

3. Student training setup

Training setting	Value
Base model	Qwen/Qwen3-1.7B
Training objective	Supervised fine-tuning on teacher-generated reasoning traces
Accepted train / val	1,922 / 485
Epochs	3
Learning rate	2×10^{-4}
Maximum sequence length	1,536 tokens
Per-device batch size	1
Gradient accumulation	8
QLoRA quantization	4-bit NF4, double quantization
LoRA rank / alpha / dropout	16 / 32 / 0.05
Target modules	all-linear

Final training summary

0.1009 training loss	0.1364 validation loss	95.84% validation token accuracy	94 min training time
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Validation token accuracy measures how well the model fits teacher-generated target sequences. It should not be interpreted as GSM8K reasoning accuracy. The main evidence of task improvement comes from the separate held-out GSM8K exact-match evaluation.

4. Evaluation methodology

- Dataset: openai/gsm8k, main configuration, official test split, 1,319 examples.
- Same fixed dataset revision for before- and after-SFT student runs.
- Deterministic decoding: seed 42, temperature 0.0, top-p 1.0.
- Qwen thinking mode disabled for the reported student comparison.
- Maximum input length: 1,536 tokens; maximum new tokens: 768.
- Same prompt, answer extraction, numeric normalization, and scoring code before and after SFT.
- Primary metric: exact-match correctness of the final numerical answer.

Metric clarification

The before-SFT accuracy is 74.68% exact match. The evaluator also reports a stricter correct-and-valid composite that additionally penalizes response-format violations; that composite is not used as the baseline mathematical accuracy.

5. Primary result: mathematical accuracy

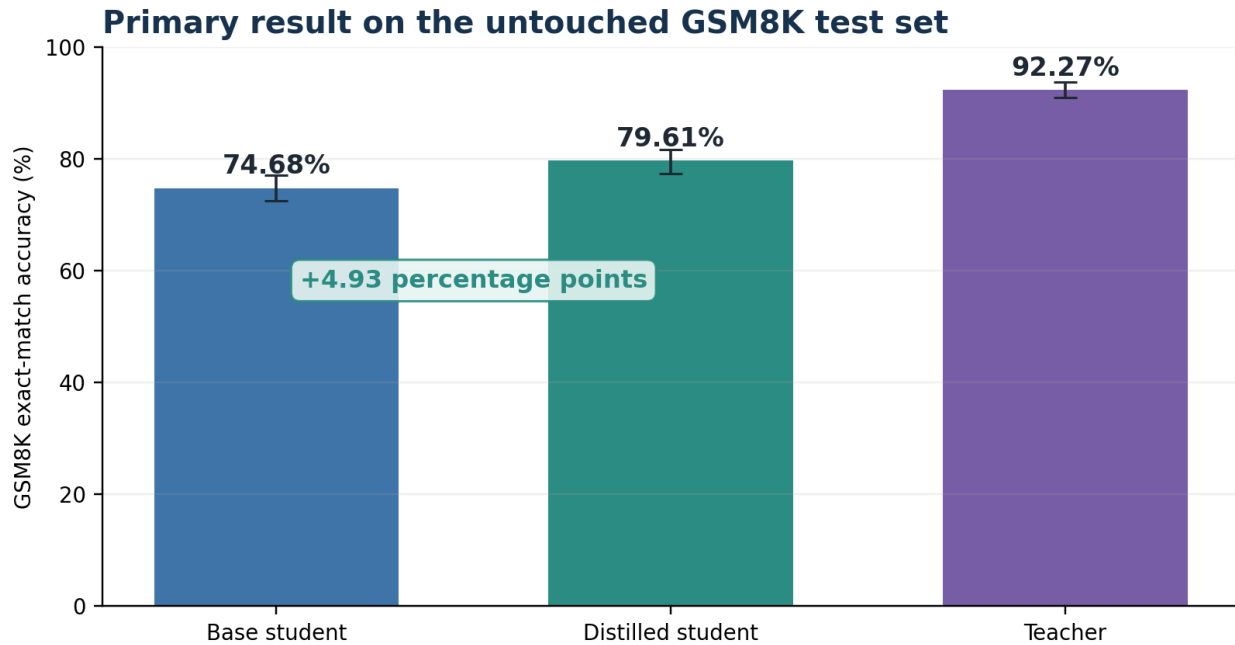


Figure 2. Exact-match accuracy with evaluator-reported 95% confidence intervals.

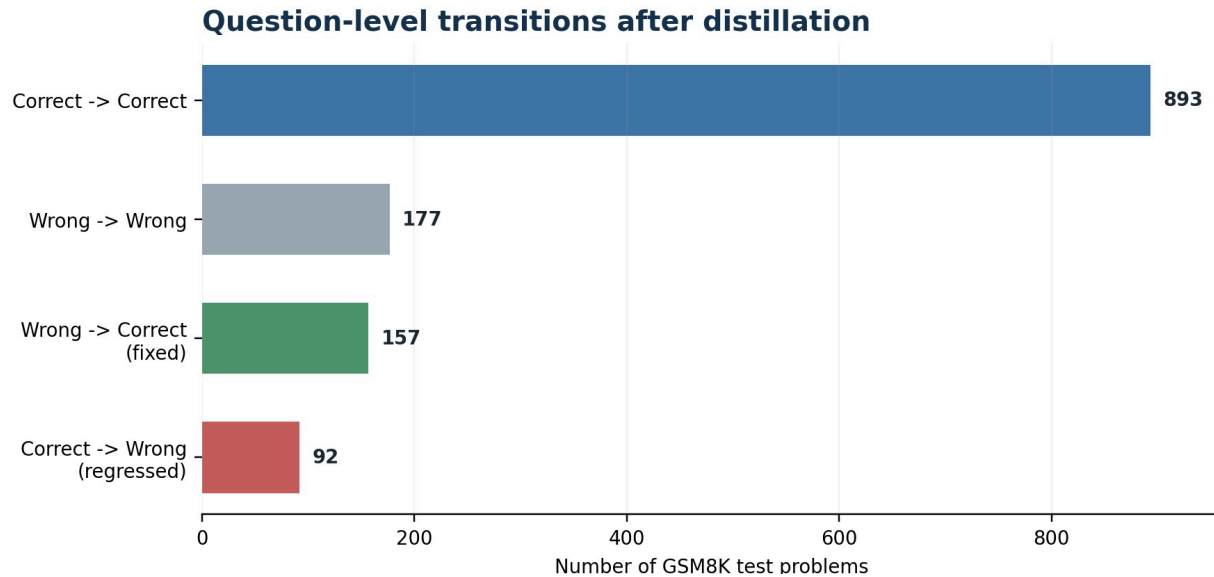
985 -> 1050 correct answers	+4.93 pp absolute gain	334 -> 269 incorrect answers	19.5% error reduction
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Before SFT, the student solved 985/1,319 test problems correctly (74.68%). After distillation, it solved 1,050/1,319 (79.61%). The evaluator-reported confidence intervals were 72.40%-77.10% before SFT and 77.33%-81.65% after SFT.

6. Question-level paired analysis

Aggregate accuracy can hide whether fine-tuning mostly fixes errors or simply swaps one set of mistakes for another. Because both models were evaluated on the same 1,319 test problems, we can inspect every before/after transition directly.

Transition	Problems
Correct -> Correct	893
Wrong -> Wrong	177
Wrong -> Correct (fixed)	157
Correct -> Wrong (regressed)	92



Question-level analysis: what changed across the 1,319 test problems

The paired view is stronger than aggregate accuracy alone because the base and distilled models answered the exact same 1,319 questions. Each problem can therefore be tracked directly from its before-SFT correctness state to its after-SFT correctness state.



Direction of the changed outcomes

A total of 249 questions changed correctness state after SFT. Of those changed outcomes, 157 improved and 92 regressed. That means 63.1% of the changed questions moved in the favorable direction, compared with 36.9% that moved in the unfavorable direction. Fine-tuning fixed about 1.71 problems for every problem that regressed.

Why the net gain matters

The difference between fixes and regressions is $157 - 92 = 65$. This exactly matches the increase from 985 to 1,050 correct answers. The +4.93-point accuracy gain is therefore traceable to a net excess of recovered problems rather than merely a reshuffling of which questions were wrong.

Paired significance test

An exact McNemar/binomial test uses only the 249 discordant pairs - the questions that changed correctness state - and tests whether fixes and regressions are equally likely. The result is $p = 4.56e-05$, providing strong evidence that the observed direction of change is favorable rather than random answer switching.

Paired evidence: 157 fixes vs. 92 regressions -> +65 net correct, exactly matching the aggregate improvement.

7. What changed after distillation?

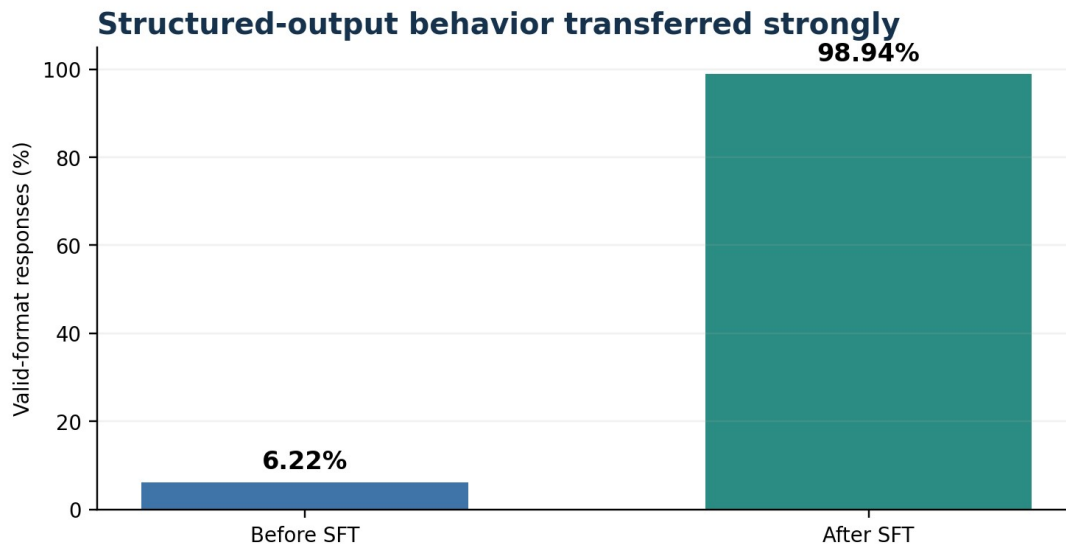
7.1 Mathematical task performance improved

Exact-match accuracy increased from 74.68% to 79.61%. Because exact match is based on the extracted numerical answer rather than perfect XML-style formatting, the +4.93-point improvement cannot be explained purely by learning the output template. The distilled model produced 65 more correct final answers on the held-out test set.

7.2 Generated solutions became more consistent with the granular reasoning style

The teacher demonstrations explicitly used small arithmetic transitions. After SFT, average completion length increased from 171.5 to 202.3 tokens, an 18.0% increase. The generated solutions therefore became more consistent with the explicit, granular style used during distillation.

7.3 Structured-output behavior transferred extremely effectively



Valid-format generation increased from 6.22% to 98.94%. The student therefore became far more reliable at producing the requested `<reasoning>` and `<final_answer>` structure with a normalized numeric final answer. This is a clear behavioral transfer from the teacher-generated supervision and is useful for downstream systems that require predictable machine-readable outputs.

What the experiment supports

Supported by the observed results

1. Mathematical task performance improved on the held-out GSM8K test set: 74.68% -> 79.61% exact match.
2. The distilled student produced 65 additional correct answers net, with 157 fixes versus 92 regressions.
3. Structured-output behavior transferred strongly: valid format increased from 6.22% to 98.94%.
4. Generated solutions became more consistent with the granular reasoning style used in the teacher demonstrations.

8. Comparison with the teacher

Model / condition	Exact match	Correct / 1,319
Qwen3-1.7B base	74.68%	985
Qwen3-1.7B distilled	79.61%	1,050
Qwen3-14B-AWQ teacher	92.27%	1,217

Distillation closed about 28% of the original teacher gap

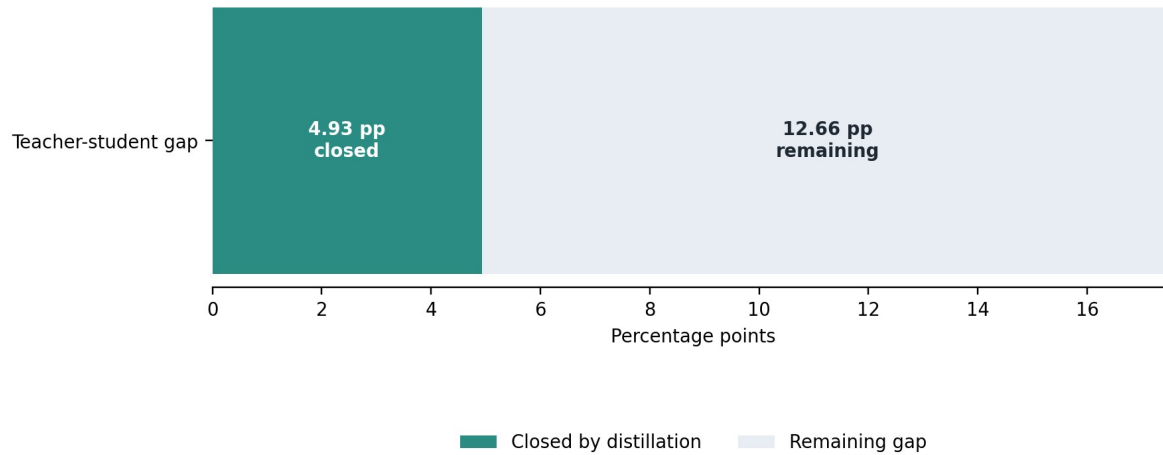


Figure 5. Portion of the original teacher-student accuracy gap closed by distillation.

The base student was 17.59 percentage points behind the teacher. After distillation, the remaining gap was 12.66 points, so the student closed about 28% of the original gap. The teacher remains substantially stronger, which is consistent with incomplete transfer and the much larger teacher capacity.

Teacher comparison caveat

The teacher run used a 2,048-token maximum generation budget, whereas student runs used 768. Teacher accuracy is therefore best treated as a reference point rather than a perfectly generation-budget-matched comparison. The before-vs-after student comparison itself is matched.

9. Efficiency trade-off

Metric	Before SFT	After SFT
Average completion tokens	171.5	202.3
Average inference time / example	14.51 s	32.12 s
Completion throughput	11.82 tok/s	6.30 tok/s
Peak CUDA memory	3.34 GiB	3.41 GiB

Distillation improved accuracy but increased generation cost in this evaluation. Outputs became about 18.0% longer, while average latency increased substantially. Peak GPU memory changed only modestly. Because the before- and after-SFT models generate different output distributions, latency should be interpreted as an observed deployment trade-off rather than a pure model-compute comparison.

10. Main scientific interpretation

The experiment provides evidence that a small language model can acquire measurable additional mathematical problem-solving capability from a larger teacher when supervised on carefully structured reasoning demonstrations.

The improvement cannot be explained simply by learning the teacher's response format. Three changes occurred simultaneously:

1. Final-answer accuracy improved	74.68% -> 79.61%
2. More failures were corrected than successes were lost	157 fixes vs. 92 regressions
3. Structured reasoning behavior transferred strongly	6.22% -> 98.94% valid format

Taken together, these results support measurable transfer of both mathematical task competence and structured output behavior from teacher-generated supervision. The distilled student remains below the teacher at 79.61% versus 92.27%, so the transfer is meaningful but partial.

11. Scope and evaluation notes

- Held-out evaluation: all final student results use the untouched official GSM8K test split; test questions were not used as student supervision.
- Same-domain scope: both distillation data and evaluation come from GSM8K, so the reported result demonstrates held-out generalization within this domain.
- Single final run: the reported student result is from the final QLoRA training run rather than an average across multiple training seeds.
- Teacher reference: the teacher used a 2,048-token generation budget versus 768 for the student runs; the base-vs.-distilled student comparison is fully matched.

12. Conclusion

Bottom line

Qwen3-1.7B improved from 74.68% to 79.61% GSM8K exact match after knowledge distillation using automatically validated Qwen3-14B-AWQ teacher-generated reasoning traces. It solved 65 additional test problems net, reduced errors by 19.5%, and increased valid structured outputs from 6.22% to 98.94%. Overall, knowledge distillation helped Qwen3-1.7B, producing a statistically significant held-out GSM8K accuracy gain and much more consistent structured outputs.

References

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